Research Article

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Chemical composition and protective effect of guava (*Psidium guajava* L.) leaf extract on piglet intestines

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Abstract

BACKGROUND: Dietary intervention is an important approach to improve intestinal function of weaned piglets. Phytogenic and herbal products have received increasing attention as in-feed antibiotic alternatives. This study investigated the chemical composition of guava leaf extract (GE) by ultrahigh-performance liquid chromatography–tandem mass spectrometry (UPLC-MS/MS). Meanwhile, we investigated the effects of dietary supplementation with GE on diarrhea in relation to immune responses and intestinal health in weaned piglets challenged by enterotoxigenic *Escherichia coli* (ETEC).

RESULTS: In total, 323 characterized compounds, which including 91 phenolic compounds and 232 other compounds were identified. Animal experiment results showed that the supplementation of 50–200 mg kg^{-1} of GE in the diet could reduce diarrhea incidence, increase activities of superoxide dismutase, glutathione peroxidase and total anti-oxidant capacity in the serum ($P < 0.05$), decrease the levels of interleukin 1β, interleukin 6 and tumor necrosis factor α in the serum or jejunum mucosa ($P < 0.05$), and increase villus height and villus height to crypt depth ratio ($P < 0.05$) in the jejuna of piglets challenged by oral ETEC compared with negative control group (NC). Meanwhile, diet supplementation with 50–200 mg kg^{-1} GE reduced the levels of D-lactate, endothelin-1 and diamine oxidase in the serum, and increased the expression of zonula occludens-1, Claudin-1, Occludin and Na^+/H^+ exchanger 3 ($P < 0.05$) in the jejuna mucosa of piglets challenged by ETEC compared with the NC.

CONCLUSIONS: These results suggested that GE could attenuate diarrhea and improve intestinal barrier function of piglets challenged by ETEC.

Supporting information may be found in the online version of this article.

Keywords: weaned piglets; enterotoxigenic *Escherichia coli*; diarrhea; inflammation; intestinal barrier

INTRODUCTION

In the swine industry, weaning period is one of the most stressful phases. Weaning stress is often associated with impairment of mucosal barrier and disruption of micro ecological balance in the small intestine of piglets, which affect the digestion and absorption of nutrients.[1,2] Once the intestinal structure and function is damaged, the potential pathogens in the intestinal tract will invade and colonize, thus causing various diseases, especially inflammation and diarrhea. Among the potential pathogens, *Escherichia coli* are the most common causative agent of post-weaning gut inflammation and diarrhea.[3] Therefore, it is important to prevent gut inflammation and maintain normal barrier function of weaned piglets.

Dietary intervention is a viable and practical approach to relieve intestinal barrier dysfunction of weaned piglets. Even through, antibiotics have been found effective in improvement of intestinal health. However, their inappropriate, irregular and continuous use may lead to drug resistance and residues. Therefore, the identification or development of alternatives to sub-therapeutic antibiotics for use in weaned piglets were attracted more research interest. Phytogenic and herbal products have received increasing attention used as in-feed antibiotic alternatives. Among the potential antibiotic alternatives in the feed, plant extracts have shown their beneficial effects on the growth performance, antioxidant capacity, and gut immunity of livestock.[4,5]

Guava (*Psidium guajava*) is a phytotherapic plant, which leaf has been widely used in folk medicine as a common traditional medicine that is believed to have active components that help to treat various diseases, such as acute and chronic enteritis, dysentery, diarrhea, etc.[6] Previous studies have partially reported composition and biological activities of guava leaf, including antioxidant,[7] antimicrobial,[8] anti-inflammatory, antibacterial[9] and anti-diarrheal[10] activities in model animal or *in vitro*. The significant biological properties of guava leaf render it a potential alternative for in-feed antibiotics.

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However, little is known about the more detailed chemical composition of the guava leaf extract (GE). And the beneficial effects of GE on the enhancement of the intestinal immune function in weaned piglets have rarely been reported. Thus, in this study, we investigated the chemical composition of GE by ultrahigh-performance liquid chromatography–tandem mass spectrometry (UPLC-MS/MS), and investigated the effects of dietary supplementation with GE on intestinal immune function in weaned piglets challenged by enterotoxigenic *Escherichia coli* (ETEC).

MATERIAL AND METHODS

Preparation of guava leaf extract (GE)

In the guava plantations, guava stem leaf is usually pruning after fruit harvest. We collected the fresh guava leaf from the guava plantations during the pruning period in July at Qionghai city of Hainan, China. Fresh leaves were oven-dried at 60 °C for 24 h and pulverized with a plant grinder. The leaf powder (50 kg) was extracted using the volume fraction of 95% ethanol at room temperature for three times, and 7 days every time. Ethanol crude extract of guava leaf was concentrated under reduced pressure by rotary evaporator to obtain ethanol extract (3.47 kg). Then, one part of obtained extract was dissolved in methanol for composition analysis, and another part of obtained extracts of guava leaf were stored at 4 °C for animal experiment.

UPLC-MS/MS qualitative analysis of GE

UPLC-MS/MS analyses were performed using an UPLC system (Dionex™ UltiMate™ 3000, Thermo Scientific) coupled to Q Exactive plus mass spectrometer (Orbitrap MS, Thermo Scientific). The chromatographic assays were performed on ZORBAX RRHD Eclipse XDB-C18 column (2.1 mm × 100 mm, 1.8 μm particle size, Agilent Technologies, USA), with a fixed temperature at 40 °C, mobile water phases with 0.1% formic acid (A) and acetonitrile (B). The elution gradient was set as follows: 0 ~ 1.0 min, 5% B; 1.0 ~ 10.0 min, 5% ~ 50% B; 10.0 ~ 16.0 min, 50–95% B; 16.0 ~ 21.0 min, 95%; 21.1 ~ 25.0 min, 5% B. The flow rate was 0.35 mL min^{-1}. The injected volume was 2 μL. The QE plus mass spectrometer was used for its ability to acquire MS/MS spectra on information-dependent acquisition (IDA) mode in the control of the acquisition software (Xcalibur, Thermo Scientific). In this mode, the acquisition software continuously evaluates the full scan MS/ddms^2 spectrum. The HESI source conditions were set as following: sheath gas flow rate as 45 arb, aux gas flow rate as 15 arb, capillary temperature as 320 ° C, full MS scan range as 100 to 1200 *m/z*, full MS resolution as 70 000, MS/MS resolution as 17 500, collision energy as 30 October 1950 in NCE mode, spray voltage as 3.8 kV (positive) or −3.0 kV (negative), respectively. The accurate-mass capability of the MS analyzer allowed reliable confirmation of the identity of the detected metabolites, normally with mass errors bellowed ±10 ppm in routine analysis, which was sufficient to verify the elemental compositions of the major constituents in GE.

Data acquisition and analysis were performed using Xcaliber 4.1 and Compound Discoverer 3.1 software (Thermo Fisher Scientific Inc., USA) and online databases: mzCloud (https://www.mzcloud.org).

Effect of GE on the intestine function in weaned piglets

Animals and experimental design

All animal experiments were conducted in accordance with the Laboratory Animal Requirements of Environment and Housing Facilities.[11] The study protocol was approved by the Institutional Animal Care and Use Committee of the Chinese Academy of Tropical Agricultural Sciences (Haikou, China).

A total of sixty weaned piglets (Duroc × Yorkshire × Landrace), with an average weight of 7.35 ± 0.18 kg (21 ± 3 days of age), were randomly assigned to six groups with five replicate pens per group (two piglets per pen). The trial lasted 28 days and consisted of six groups. The groups were divided as follows: (i) blank control group (BC), piglets without ETEC challenge were fed diet without supplements; (ii) negative control group (NC), piglets were fed diet without supplements and challenged by ETEC; (iii) positive control group (PC), piglets were fed diet supplemented with 50 mg kg^{-1} quinocetone and challenged by ETEC; (iv) T50 group (T50), piglets were fed diet supplemented with 50 mg kg^{-1} GE and challenged by ETEC; (v) T100 group (T100), piglets were fed diet supplemented with 100 mg kg^{-1} GE and challenged by ETEC; (vi) T200 group (T200), piglets were fed diet supplemented with 200 mg kg^{-1} GE and challenged by ETEC. The diet was formulated to base on the nutritional requirements of the National Research Council (NRC, 2012), and the ingredients and composition of the diet are presented in Table 1. Quinocetone was purchased from Dalian Ronghai Biotechnology Co., Ltd. (China), and used as antibiotic control.[12,13]

Throughout the 28 day's study, all pigs had *ad libitum* access to feed and water. On the day 4, all piglets (except for in the blank control group) were orally challenged by 10^9 colony-forming units (CFU) enterotoxigenic *E. coli* (ETEC) according to the method developed by Wu *et al.*[14] ETEC was obtained from China Veterinary Culture Collection Center (Beijing, China). The animals were kept in a temperature-controlled environment at 25 °C and

Table 1. Composition and nutrient levels of diets (as-fed basis)

Ingredient	Content, g kg^{-1}	Nutrient levels, g kg^{-1c}	
Corn	579.0	DE, MJ kg^{-1}	14.02
Soybean meal	254.7	Crude protein	200.5
Fish meal	50.0	Ca	6.2
Whey powder	40.0	Available P	5.0
Cream powder	45.0	Lys	15.9
Limestone	3.0	Met	5.6
$Ca(H_2PO_4)_2$	12.0	Met+Cys	8.5
Moldproofant	1.0		
Acidifier	3.0		
L-Lysiine·HCl	2.5		
Choline chloride	1.0		
DL-Methionine	0.5		
NaCl	3.0		
Trace mineral premix[a]	5.0		
Vitamin premix[b]	0.3		
Total	1000.0		

[a] Trace mineral premix provides the following per kilogram of diets: 16.5 mg of Cu as $CuSO_4{\cdot}5H_2O$, 100 mg of Fe as $FeSO_4{\cdot}H_2O$, 35 mg of Mn as $MnSO_4{\cdot}H_2O$, 100 mg of Zn as $ZnSO_4{\cdot}H_2O$, 0.3 mg of Se as Na_2SeO_3, 0.3 mg of I as KI.
[b] Vitamin premix provides the following per kilogram of diets: 11 000 IU of vitamin A, 1500 IU of vitamin D3, 16 IU of vitamin E, 1 mg of vitamin K, 0.3 mg of folic acid, 15 mg of nicotinic, 10 mg of pantothenate, 2.5 mg of biotin, 1 mg of vitamin B1, 4.0 mg of vitamin B2, 0.01 mg of vitamin B12.
[c] Nutrient levels were calculated values.

Table 2. Identification of phenolics in GE by UPLC-MS/MS

No.	Retention time (min)	Compound	Molecular formula	Adducts[a]	Error[b] (ppm)	Score[c]
Flavonoids						
1	2.131	gallocatechin	$C_{15}H_{14}O_7$	[M+H]	1.95	98.7
2	2.327	(-)-epigallocatechin	$C_{15}H_{14}O_7$	[M+H]	1.95	98.1
3	3.780	catechin	$C_{15}H_{14}O_6$	[M-H]	1.70	96.7
4	3.803	epicatechin	$C_{15}H_{14}O_6$	[M+H]	2.13	98.7
5	4.911	myricetin 3-*O*-*β*-*D*-galactopyranoside	$C_{21}H_{20}O_{13}$	[M+H]	1.60	98.6
6	4.934	quercetin-3*β*-*D*-glucoside	$C_{21}H_{20}O_{12}$	[M+H]	1.53	98.9
7	4.994	quercetin-3-*O*-vicianoside	$C_{26}H_{28}O_{16}$	[M-H]	1.78	91.0
8	5.126	kaempferol 3-gentiobioside	$C_{27}H_{30}O_{16}$	[M+H]	1.62	99.8
9	5.243	naringeninchalcone	$C_{15}H_{12}O_5$	[M+H]	1.90	93.7
10	5.289	rutin	$C_{27}H_{30}O_{16}$	[M-H]	1.58	98.3
11	5.403	pueraria glycoside	$C_{21}H_{20}O_{10}$	[M+H]	1.43	96.5
12	5.468	kaempferol 3- *O*-*β*-*D*-arabinopyranosyl-*β*-*D*-glucopyranoside	$C_{20}H_{18}O_{15}$	[M+H]	1.81	99.8
13	5.491	miquelianin	$C_{21}H_{18}O_{13}$	[M+H]	4.05	99.4
14	5.505	hyperoside	$C_{21}H_{20}O_{12}$	[M+H]	1.33	99.5
15	5.729	quercetin-*α*-*L*-arabinose	$C_{26}H_{28}O_{11}$	[M+H]	1.95	99.6
16	5.774	kaempferol 3-neohesperidoside	$C_{27}H_{30}O_{15}$	[M+H]	1.80	99.5
17	5.834	astragalin	$C_{21}H_{20}O_{11}$	[M-H]	1.79	95.2
18	5.855	trifolin	$C_{21}H_{20}O_{11}$	[M+H]	1.80	99.5
19	5.931	avicularin	$C_{20}H_{18}O_{11}$	[M+H]	1.95	99.5
20	5.937	morin	$C_{15}H_{10}O_7$	[M+H]	2.21	97.0
21	6.011	kaempferol-3-*O*-glucuronoside	$C_{21}H_{18}O_{12}$	[M+H]	2.46	98.9
22	6.136	isorhamnetin-3-*O*-glucoside	$C_{22}H_{22}O_{12}$	[M+H]	1.65	98.2
23	6.220	puerarin	$C_{21}H_{20}O_9$	[M+H]	1.68	98.9
24	6.402	myricetin	$C_{15}H_{10}O_8$	[M+H]	1.88	94.5
25	6.479	juglanin	$C_{20}H_{18}O_{10}$	[M+H]	2.29	99.0
26	6.538	phloretin	$C_{15}H_{14}O_5$	[M+H]	1.53	99.6
27	6.544	maesopsin	$C_{15}H_{12}O_6$	[M-H]	1.57	90.8
28	6.679	quercetin 3-methyl ether 3′-xyloside	$C_{21}H_{20}O_{11}$	[M+H]	1.80	85.3
29	6.944	chrysin 8-*C*-glucoside	$C_{21}H_{20}O_9$	[M+H]	1.68	86.6
30	7.584	quercetin	$C_{15}H_{10}O_7$	[M+H]	1.42	98.6
31	8.162	pinoquercetin	$C_{16}H_{12}O_7$	[M+H]	1.23	98.5
32	8.422	naringenin	$C_{15}H_{12}O_5$	[M+H]	2.34	95.9
33	8.433	genistein	$C_{15}H_{10}O_5$	[M-H]	1.19	88.4
34	8.629	kaempferol	$C_{15}H_{10}O_6$	[M+H]	0.94	99.5
35	8.687	diosmetin	$C_{16}H_{12}O_6$	[M+H]	1.49	97.9
36	8.849	isorhamnetin	$C_{16}H_{12}O_7$	[M+H]	1.14	99.5
37	8.998	apigenin	$C_{15}H_{10}O_5$	[M-H]	1.19	92.2
38	9.084	poriol	$C_{16}H_{14}O_5$	[M-H]	1.40	88.9
39	9.295	kaempferide	$C_{16}H_{12}O_6$	[M+H]	1.49	95.2
40	10.680	chrysin	$C_{15}H_{10}O_4$	[M+H]	1.61	92.4
41	11.738	formononetin	$C_{16}H_{12}O_4$	[M+H]	1.67	94.5
42	11.904	pinostrobin	$C_{16}H_{14}O_4$	[M+H]	1.77	96.8
43	14.843	galangin	$C_{15}H_{10}O_5$	[M+H]	1.84	95.3
Coumarins						
44	4.618	magnolioside	$C_{16}H_{18}O_9$	[M+H]	1.66	98.3
45	4.863	esculetin	$C_9H_6O_4$	[M+H]	1.01	93.4
46	5.431	4-hydroxycoumarin	$C_9H_6O_3$	[M-H]	1.99	97.7
47	5.458	7-hydroxycoumarine	$C_9H_6O_3$	[M+H]	1.66	92.9
48	5.564	scopoletin	$C_8H_{10}O_4$	[M+H]	−0.78	86.3
49	6.854	4-hydroxycoumarin	$C_9H_6O_3$	[M+H]	1.66	87.9
50	7.011	5,7-dihydroxy-4-methylcoumarin	$C_{10}H_8O_4$	[M-H]	1.36	98.7
51	11.948	ostruthin	$C_{19}H_{22}O_3$	[M-H]	−8.99	87.6
52	12.509	8-hydroxy-7-methoxy-3-(2-methylbut-3-en-2-yl)-2*H*-chromen-2-one	$C_{15}H_{16}O_4$	[M+Na]	−7.49	96.6
Cinnamic acids						
53	3.915	ferulic acid 1-*O*-glucoside	$C_{15}H_{18}O_8$	$[M+NH_4]$	1.54	98.3
54	4.180	caffeic acid	$C_9H_8O_4$	[M-H]	1.79	92.1

Table 2. Continued

No.	Retention time (min)	Compound	Molecular formula	Adducts[a]	Error[b] (ppm)	Score[c]
55	6.635	3-(2-glucosyloxy-4-methoxyphenyl) propanoic acid	$C_{16}H_{22}O_9$	[M-H]	1.65	85.6
56	6.686	*DL*-4-hydroxyphenyllactic acid	$C_9H_{10}O_4$	[M-H]	1.38	88.2
57	8.211	ferulic acid	$C_{10}H_{10}O_4$	[M-H]	1.71	85.8
Phenolic acids and aldehydes						
58	1.245	gallic acid	$C_7H_6O_5$	[M-H]	2.31	99.5
59	2.079	gentisic acid 5-*β*-glucoside	$C_{13}H_{16}O_9$	[M-H]	1.59	90.6
60	2.187	6-methoxysalicylic acid	$C_8H_8O_4$	[M-H]	2.27	97.8
61	2.851	4-methoxysalicylic acid	$C_8H_8O_4$	[M-H]	2.33	98.8
62	4.103	resorcinol monoacetate	$C_8H_8O_3$	[M-H]	2.18	99.1
63	4.176	gentisic acid	$C_7H_6O_4$	[M-H]	1.83	97.9
64	5.617	syringic acid	$C_9H_{10}O_5$	[M-H]	1.98	90.3
65	6.544	4-hydroxybenzoic acid	$C_7H_6O_3$	[M-H]	2.19	98.0
66	11.603	4-hydroxy-3-[(5*E*)-7-hydroxy-3,7-dimethyl-4-oxo-5-octen-1-yl]-5-[(2*E*)-4-hydroxy-3-methyl-2-buten-1-yl] benzoic acid	$C_{22}H_{30}O_6$	[M-H]	1.13	85.9
67	13.560	erionic acid E	$C_{22}H_{30}O_5$	[M-H]	1.61	86.3
Simplie phenolics						
68	1.495	maltol	$C_6H_6O_3$	[M+H]	1.34	99.0
69	2.558	catechol	$C_6H_6O_2$	[M-H]	3.03	99.0
70	2.577	2-[(4-hydroxy-3,5-dimethoxyphenyl) methoxy]-6-(hydroxymethyl) oxane-3,4,5-triol	$C_{15}H_{22}O_9$	[M+NH4]	2.20	98.8
71	3.289	salidroside	$C_{14}H_{20}O_7$	[M+NH4]	1.70	92.0
72	3.611	osmanthuside H	$C_{19}H_{28}O_{11}$	[M+H]	2.24	98.1
73	4.039	koaburside	$C_{15}H_{22}O_9$	[M+H]	1.76	98.4
74	5.563	4-hydroxy-3,5,6-trimethyl-2*H*-pyran-2-one	$C_8H_{10}O_3$	[M+H]	1.29	92.6
75	7.278	2,3,6-trimethylphenol	$C_9H_{12}O$	[M-H]	1.55	92.0
76	10.220	BHQ	$C_{14}H_{22}O_2$	[M-H]	1.58	85.6
77	11.065	3-tert-butylphenol	$C_{10}H_{14}O$	[M-H]	2.55	97.2
78	14.127	3,5-di-tert-butyl-4-hydroxybenzyl alcohol	$C_{15}H_{24}O_2$	[M-H]	2.21	97.0
Other phenolics						
79	1.244	1,6-*bis*-*O*-(3,4,5-trihydroxybenzoyl) hexopyranose	$C_{20}H_{20}O_{14}$	[M+H-H2O]	1.93	98.5
80	3.021	corilagin	$C_{27}H_{22}O_{18}$	[M+NH4]	1.86	97.5
81	4.493	4-hydroxybenzaldehyde	$C_7H_6O_2$	[M+H]	1.63	91.4
82	4.752	1-*O*-(3-hydroxy-5-methylphenyl)-*β*-*D*-glucopyranose 6-(3,4,5-trihydroxybenzoate)	$C_{20}H_{22}O_{11}$	[M-H]	1.69	92.4
83	6.233	acanthoside B	$C_{28}H_{36}O_{13}$	[M+NH4]	1.87	95.1
84	6.581	(3,4,5-trihydroxy-6-{[4-(2,6,6-trimethyl-4-oxocyclohex-2-en-1-yl) butan-2-yl] oxy} oxan-2-yl) methyl 3,4,5-trihydroxyben zoate	$C_{26}H_{36}O_{11}$	[M+H]	1.52	95.1
85	6.827	3-hydroxymandelic acid	$C_8H_8O_4$	[M-H]	2.33	92.2
86	7.307	citrinin	$C_{13}H_{14}O_5$	[M-H]	1.73	90.0
87	7.997	clemaphenol A	$C_{20}H_{22}O_6$	[M+H-H2O]	1.29	90.0
88	8.082	ethyl paraben	$C_9H_{10}O_3$	[M-H]	2.12	94.6
89	10.396	piceatannol	$C_{14}H_{12}O_4$	[M-H]	2.06	94.1
90	10.621	1-(2,4-dihydroxyphenyl)-2-(3,5-dimethoxyphenyl) propan-1-one	$C_{17}H_{18}O_5$	[M+Na]	−5.57	97.1
91	20.418	emodin	$C_{15}H_{10}O_5$	[M+H]	6.23	97.4

[a] In the adducts column, [M+H], [M+H-H2O], [M+Na] and [M+NH4] means that compounds were detected in positive ion mode. [M-H] means that compounds were detected in negative ion mode.
[b] The values were accurate molecular weight after addition.
[c] The values were the degree of matching between the analytical compounds of the typical total ion chromatogram (Full-MS/ddMS^2 mode) of GE by the software Compound Discoverer 3.1 and the mass spectrometry of first-order and second-order of online database (https://www.mzcloud.org) in positive/negative ion mode, respectively. The values were greater than or equal to 85.0.

alternating light and dark cycles with 12 h intervals. Individual piglet body weight (BW) was measured on days 1 and days 29 to calculate average daily gain (ADG) for each pen. Average daily feed intake (ADFI) was measured per pen from the difference between the sums of feed additions and feed remaining at the end of trial. Feed efficiency (F/G) was calculated by dividing ADFI

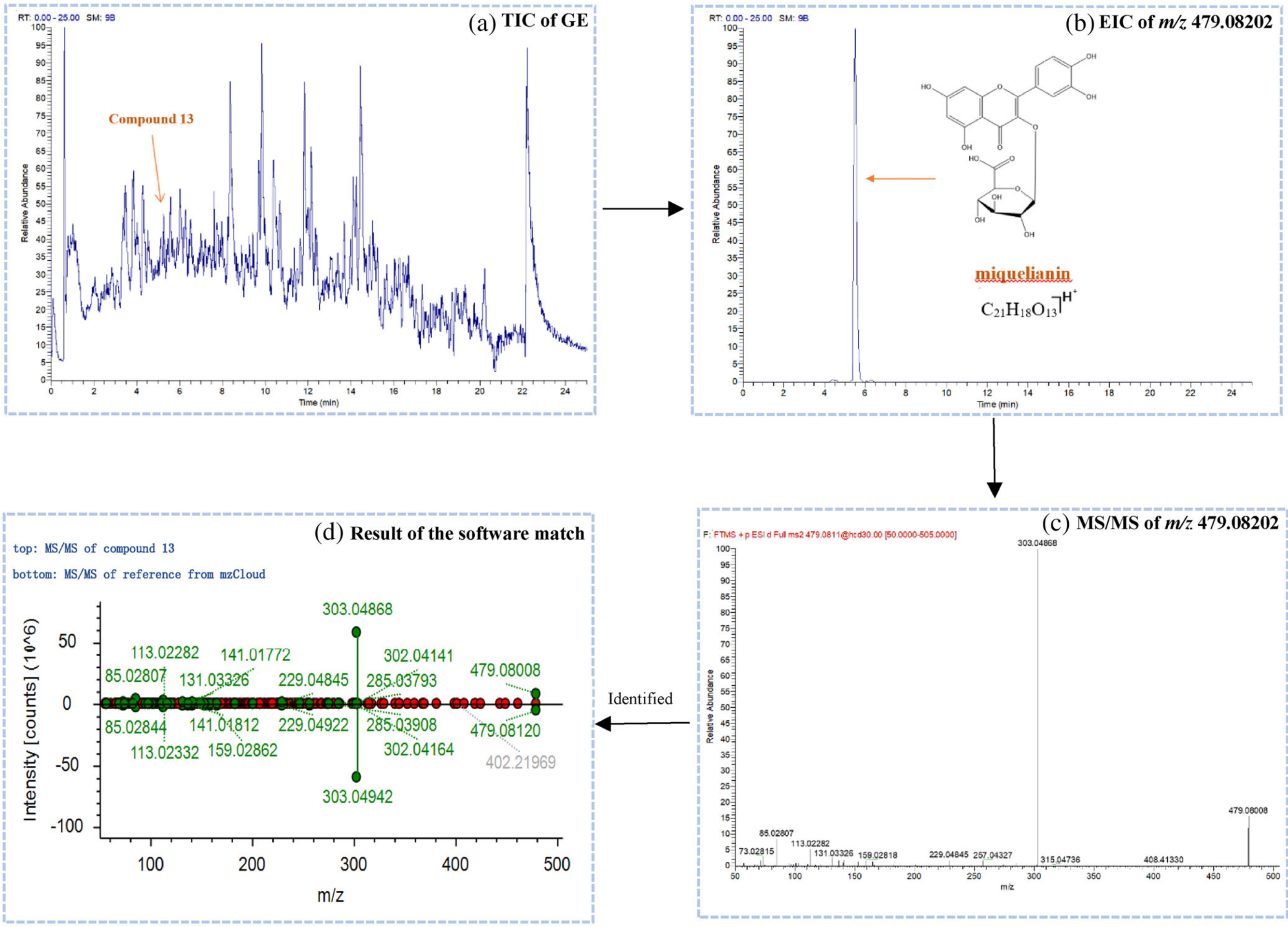

Figure 1. Screening and identifying flowchart of compound **13** by Compound Discoverer 3.1 software.

Table 3. Effect of GE on the growth performance of piglets

Items	Groups BC	NC	PC	T50	T100	T200	SEM	*P*	Contrast[a] *L*	*Q*
Initial weight, kg	7.31	7.34	7.41	7.17	7.31	7.49	0.59	0.958	0.946	0.852
Final weight, kg	14.12	13.88	14.51	15.20	13.87	13.95	0.93	0.770	0.711	0.606
ADFI, g	457	457	448	456	457	445	2	0.483	0.161	0.458
ADG, g	244	236	257	301	238	231	8	0.094	0.242	0.175
F/G	1.87	1.94	1.78	1.55	1.92	1.93	0.05	0.111	0.404	0.242
Diarrhea incidence, %	1.79	21.43	16.07	14.29	8.93	7.14	—	—	—	—

In the same row, values with no letter or the same letter superscripts mean no significant difference ($P > 0.05$), while with different letter superscripts mean significant difference ($P < 0.05$). n = 5. ADFI, average daily feed intake; ADG, average daily gain; F/G, ADFI/ADG; BC, blank control group, piglets without ETEC challenge were fed diet without supplements; NC, negative control group, piglets were fed diet without supplements and challenged by ETEC; PC, positive control group, piglets were fed diet supplemented with 50 mg kg^{-1} quinocetone and challenged by ETEC; T50, T50 group, piglets were fed diet supplemented with 50 mg kg^{-1} GE and challenged by ETEC; T100, T100 group, piglets were fed diet supplemented with 100 mg kg^{-1} GE and challenged by ETEC; T200, T200 group, piglets were fed diet supplemented with 200 mg kg^{-1} GE and challenged by ETEC.
[a] L, Linear; Q, Quadratic; Linear and quadratic effect of adding GE compared to the NC.

by ADG. The occurrence of diarrhea during the whole experiment for each group was assessed as described by Wang *et al.*,[5] and calculated based on each group.

Sample collection

At the morning of the 29th day of the experiment, one pig was randomly selected from each pen and the blood samples were collected from the jugular vein, and serum was separated by centrifugation at 700 × *g* for 15 min at 4 °C. Then the serum was aliquoted into 2 mL capacity and stored at −80 °C for further analysis. After blood sampling, pigs were anesthetized by injection of sodium pentobarbital solution (50 mg kg^{-1} of BW) and were killed by exsanguination. The small intestine was removed and a piece (4 cm-length) of the middle jejunum was collected, gently rinsed with 0.1 M phosphate buffered saline (PBS) at pH 7.2, and then fixed in 10% formaldehyde-phosphate buffer for subsequent histological analysis. About 3 g of jejunal mucosa was snap-frozen in liquid nitrogen, and stored at −80 °C for total RNA and protein isolation.

Blood sample analysis

The levels of glutathione peroxidase (GSH-Px), superoxide dismutase (SOD), total anti-oxidant capacity (T-AOC), malondialdehyde (MDA), D-lactate (D-LA), endothelin-1 (ET-1) and diamine oxidase (DAO) in the serum were measured using commercial kits according to the instructions of the manufacturer (Nanjing Jiancheng Institute of Bioengineering, Nanjing, China). The concentrations of tumor necrosis factor-α (TNF-α), interleukin-1β (IL-1β) and interleukin-6 (IL-6) in the serum were determined with swine-specific enzyme-linked immunosorbent assays (ELISAs) kits, according to the recommendations of the manufacturer (R&D Systems Inc., Minneapolis, MN, USA).

Jejunal morphology and histology analysis

The fixed intestinal samples were dehydrated in an ascending series of ethanol, cleared in xylene, and embedded in paraffin. Histological slides of 5 μm thickness were stained with hematoxylin and eosin (HE) for light microscopic examination. A total of 10 well-oriented, intact villus height and crypt depth measurements were selected and measured for each intestinal cross-section, and the means of these measurements were calculated for each sample.

RNA extraction and determination of mRNA expression by RT-PCR

Total RNA from the jejunal mucosa was extracted using TRIzol reagent (Invitrogen Corp., Carlsbad, CA, USA). RNA quality and quantity were determined with the RNA 6000 Nano Lab Chip kit in a bioanalyzer (Agilent Technologies, Shanghai, China). The integrity of RNA was determined by agarose gel (1%) electrophoresis. Total RNA (2 μg) was used to generate cDNA using a PrimeScript RT Reagent Kit (Takara, Dalian, China). Real-time PCR was performed with SYBR Green PCR master mix (Fermentas, Burlington, Canada). The primers of GAPDH, TNF-α, IL-1β, IL-6, ZO-1, Claudin-1, Occludin and NHE3 are showed in Table S1. The GAPDH gene was used as an internal control. The reaction was performed at 95 °C for 2 m; 40 cycles of 95 °C for 15 s, 59 °C for 20 s, 72 °C for 30 s; and a following cycle of 60 °C for 30 s and 95 °C for 10 s. The specificity of PCR product was assessed by melting curve analysis. All reaction was run in triplicate. Results (fold changes) were

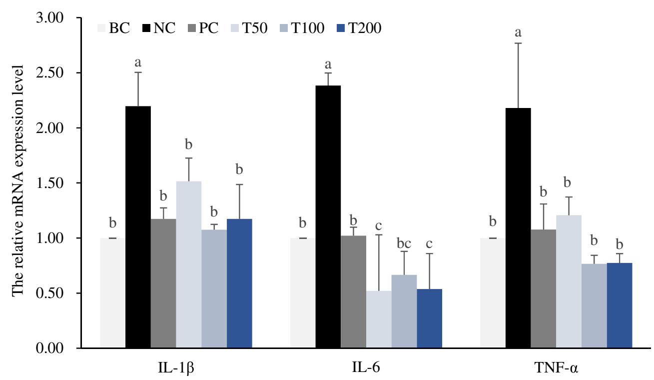

Figure 2. Effect of GE on cytokines mRNA expression in the jejunum mucosa of piglets. Values are expressed as means ± SD (n = 5). Values with no common superscripts means significant difference ($P < 0.05$). Dietary supplementation with GE linearly decreased mRNA expression of TNF-α ($P < 0.01$), IL-1β ($P < 0.01$) and IL-6 ($P < 0.01$) in the jejunum mucosa. BC, blank control group, piglets without ETEC challenge were fed diet without supplements; NC, negative control group, piglets were fed diet without supplements and challenged by ETEC; PC, positive control group, piglets were fed diet supplemented with 50 mg kg^{-1} quinocetone and challenged by ETEC; T50, T50 group, piglets were fed diet supplemented with 50 mg kg^{-1} GE and challenged by ETEC; T100, T100 group, piglets were fed diet supplemented with 100 mg kg^{-1} GE and challenged by ETEC; T200, T200 group, piglets were fed diet supplemented with 200 mg kg^{-1} GE and challenged by ETEC.

Table 4. Effect of GE on the serum antioxidant capacities and cytokines levels in piglets

	Groups								Contrast[a]	
Items	BC	NC	PC	T50	T100	T200	SEM	*P*	*L*	*Q*
SOD, U mL^{-1}	66.8^{ab}	60.3^{b}	72.7^{a}	69.7^{ab}	74.2^{a}	70.5^{ab}	1.4	0.012	0.009	0.020
GSH-Px, U mL^{-1}	566^{ab}	330^{c}	429^{bc}	677^{a}	552^{ab}	467^{bc}	29	<0.001	0.076	<0.001
T-AOC, U mL^{-1}	1.60^{b}	0.27^{c}	0.99^{bc}	2.04^{ab}	1.42^{b}	2.78^{a}	0.22	<0.001	0.003	0.422
MDA, nmol mL^{-1}	2.30^{b}	3.22^{a}	2.30^{b}	2.53^{ab}	2.55^{ab}	2.54^{ab}	0.09	0.013	0.008	0.027
IL-1β, ng L^{-1}	18.7^{b}	30.8^{a}	8.1^{c}	15.5^{b}	16.5^{b}	10.3^{c}	1.8	<0.001	<0.001	<0.001
IL-6, ng L^{-1}	3.43^{b}	3.96^{a}	2.11^{c}	3.27^{b}	3.51^{b}	3.41^{b}	0.14	<0.001	0.002	0.002
TNF-α, ng L^{-1}	41.7^{b}	46.9^{a}	31.7^{e}	38.7^{c}	35.0^{d}	36.5^{cd}	1.2	<0.001	<0.001	<0.001

In the same row, values with no letter or the same letter superscripts mean no significant difference ($P > 0.05$), while with different letter superscripts mean significant difference ($P < 0.05$). n = 5. SOD, superoxide dismutase; GSH-Px, glutathione peroxidase; T-AOC, total anti-oxidant capacity; MDA, malondialdehyde; TNF-α, tumor necrosis factor-α; IL-1β, interleukin-1β; IL-6, interleukin-6; BC, blank control group, piglets without ETEC challenge were fed diet without supplements; NC, negative control group, piglets were fed diet without supplements and challenged by ETEC; PC, positive control group, piglets were fed diet supplemented with 50 mg kg^{-1} quinocetone and challenged by ETEC; T50, T50 group, piglets were fed diet supplemented with 50 mg kg^{-1} GE and challenged by ETEC; T100, T100 group, piglets were fed diet supplemented with 100 mg kg^{-1} GE and challenged by ETEC; T200, T200 group, piglets were fed diet supplemented with 200 mg kg^{-1} GE and challenged by ETEC.
[a] L, Linear; Q, Quadratic; Linear and quadratic effect of adding GE compared to the NC.

evaluated using the $2^{-\Delta\Delta CT}$ method with $\Delta CT = CT$ (target gene)-CT (GAPDH) and $\Delta\Delta CT = \Delta CT$ (treated group)-ΔCT (blank control group).

Western blotting

The jejunal mucosa was homogenized on ice in RIPA lysis buffer (Upstate, Temecula, CA). After centrifugation at 4 °C and 14 000 × *g*, the supernatants were collected for the assay. Polyclonal primary antibodies against GAPDH (1:5000 dilution, CMC), Claudin-1 (1:2000 dilution, Abcam), Occludin (1:2000 dilution, Abcam), ZO-1 (1:2000 dilution, Abcam) and NHE3 (1:2000 dilution, Abcam) were employed, followed by incubation with a goat anti-rabbit IgG (1:5000 dilution, ABclonal) secondary antibody. The results were normalized to the GAPDH standard and analyzed using the image analysis software, AlphaEase FC 4.0 (Alpha Innotech, Co., USA). All the immunoblotting assays were performed using three biological replicates.

Statistical analysis

Statistical analysis of growth performance, serum indexes, jejunum morphology, and gene and protein expression among the groups were evaluated by using the one-way analysis of variance (ANOVA), performed using SPSS 23.0 (IBM-SPSS Inc., Chicago, USA). The results in the tables were presented as mean ± standard error of mean (SEM), and other figure results were shown with means ± standard deviation (SD). Orthogonal polynomial contrasts were used to test the linear and quadratic effects of GE by comparing with the negative control group. Significant differences were evaluated by Tukey multiple comparisons test at $P < 0.05$.

RESULTS

Chemical composition of GE

The total ion chromatograms (TIC), corresponding to the positive and negative signals were showed in Fig. S1, respectively. Three hundred and twenty-three characterized compounds, which including 91 phenolic compounds (Table 2) and 232 other compounds (Table S2) were summarized with the relevant data, including retention time, molecular formula, adducts, error values (ppm) and score provided by the software and MS/MS fragments. Taking compound **13** as an example, the extracted ion chromatography (EIC) of theoretical mass (*m/z* 479.08202, [M + H]) of a reported compound in the TIC of GE (Fig. 1(a)) was performed. The result showed that a high-abundance peak appeared (Rt 5.491 min), and its measured MS data in positive mode was *m/z* 479.08008 (ppm 4.05) (Fig. 1(b)). The MS/MS spectrum was further obtained by the target MS/MS method. In the MS/MS spectrum of this metabolite (Fig. 1(c)), the fragment ions were produced at *m/z* 303.04868, 229.04845, 131.03326, 113.02282 and 85.02807, respectively. Those characteristic MS/MS data were consistent with the pure MS/MS spectrum of miquelianin in the mzCloud library by using of the software Compound Discoverer

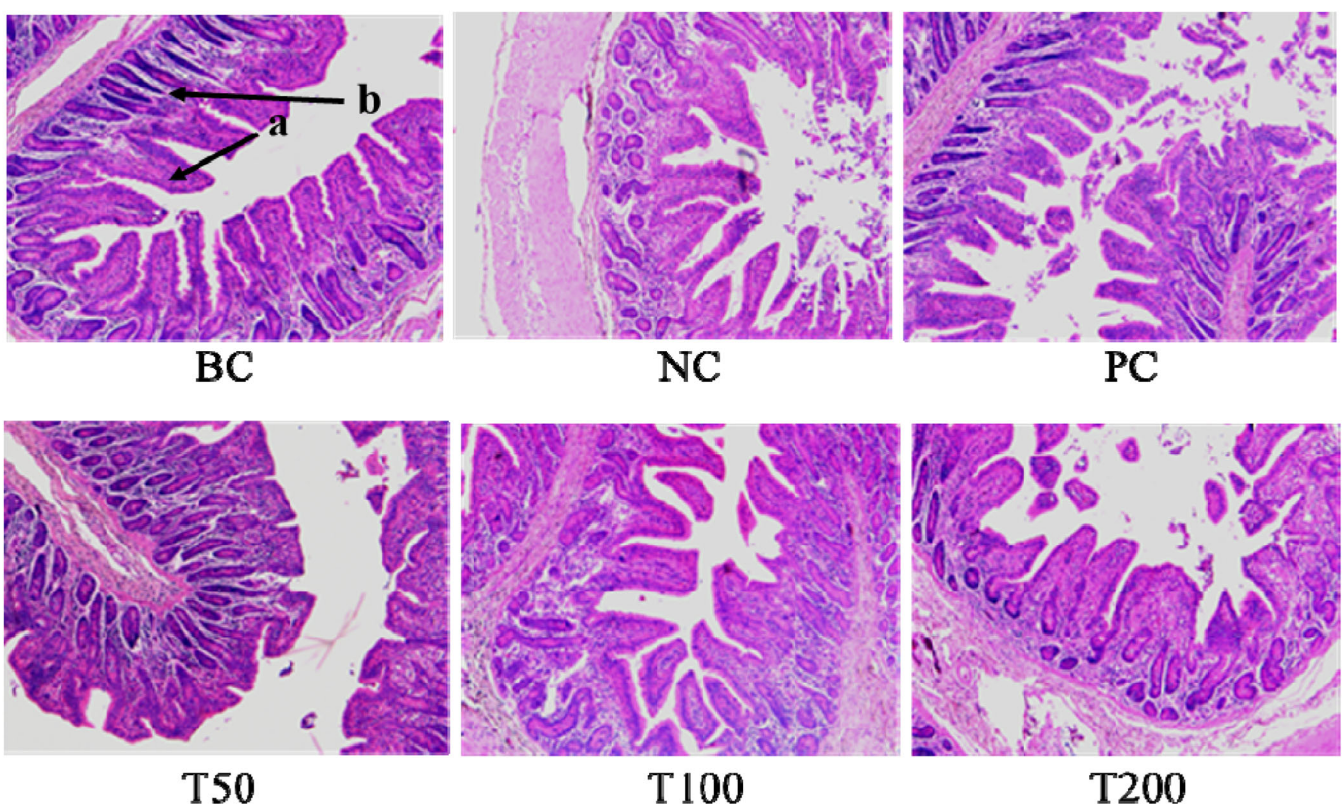

Figure 3. Effect of GE on histopathological changes of jejunum mucosa in piglets with HE staining (HE, 100×). Arrow a and b in the figure means villus and crypts, respectively. Scale bar is 100 μm. BC, blank control group, piglets without ETEC challenge were fed diet without supplements; NC, negative control group, piglets were fed diet without supplements and challenged by ETEC; PC, positive control group, piglets were fed diet supplemented with 50 mg kg^{-1} quinocetone and challenged by ETEC; T50, T50 group, piglets were fed diet supplemented with 50 mg kg^{-1} GE and challenged by ETEC; T100, T100 group, piglets were fed diet supplemented with 100 mg kg^{-1} GE and challenged by ETEC; T200, T200 group, piglets were fed diet supplemented with 200 mg kg^{-1} GE and challenged by ETEC.

3.1 for both identification and similarity searching, and the match score in the result list was 99.4. (the standard of the matched score is greater than or equal to 85.0) (Fig. 1(d)). Therefore, compound **13** was tentatively identified as miquelianin. So according to the above method, 323 compounds were detected. Phenolics were compounds that have at least one hydroxyl group attached directly to one or more aromatic rings. However, the hydrogen of the phenolic hydroxyl was labile due to the aromatic ring, which made it easier to scavenge free radicals through hydrogen or electron transferring. In Table 2, 91 phenolics were detected, including 43 flavonoids, 13 other phenols, 11 simple phenolics, 10 phenolic acids and aldehydes, 9 coumarins and 5 cinnamic acids.

Effect of dietary GE on the performance, diarrhea incidence and anti-oxidative capacities of piglets

Table 3 summarized the growth performance and diarrhea incidence of piglets. Piglets challenged by ETEC has the higher diarrhea incidence than the piglets without ETEC challenge in the BC group. The supplementation of 50 mg kg^{-1} of quinocetone, or 50–200 mg kg^{-1} of GE in diets reduced diarrhea incidence of piglets compared with the NC group. However, no difference was observed in the initial body weight, the final body weight, ADG, ADFI and F/G of piglets among groups ($P > 0.05$). No mortality was observed throughout the trial.

As shown in Table 4, ETEC challenge reduced the activities of GSH-Px and T-AOC, and increased the level of MDA in the serum of piglets compared with the BC group ($P < 0.05$). Dietary supplementation with quinocetone increased activities of SOD, and reduced the level of MDA in the serum of piglets ($P < 0.05$) compared with the NC group. Supplementation of GE in the diet linearly increased activities of SOD ($P = 0.009$) and T-AOC ($P = 0.003$), and reduced the MDA ($P = 0.008$) concentration in the serum of piglets compared with the NC group.

Effect of dietary GE on intestinal inflammation of piglets

ETEC challenge induced intestinal inflammation, and increased the pro-inflammatory factors levels of piglets in NC group, such as TNF-α, IL-1β and IL-6 ($P < 0.05$), both in the serum (Table 4) and jejunum mucosa (Fig. 2) of piglets compared with BC group. However, dietary supplementation with GE linearly decreased the levels of TNF-α ($P < 0.001$), IL-1β ($P < 0.001$) and IL-6 ($P = 0.002$) in the serum (Table 4), and mRNA expression of TNF-α ($P < 0.001$), IL-1β ($P < 0.001$) and IL-6 ($P < 0.001$) in the jejunum mucosa (Fig. 2) of piglets compared with the NC group.

Effect of dietary GE on intestinal morphology of piglets

We can see in Fig. 3, ETEC challenge caused intestinal mucosal damage in piglets, such as reduced villus height, villus loss, compared with the BC group. However, dietary supplementation with quinocetone or 50–200 mg kg^{-1} GE alleviated the damage of intestinal mucosa in piglets compared with the NC group. As shown in Table 5, compared with the BC group, ETEC challenge increased crypt depth ($P < 0.05$) and decreased both villus height and villus height to crypt depth ratio ($P < 0.05$) in the jejunum of piglets

Table 5. Effect of GE on jejunum mucosa structure of piglets

	Groups								Contrast[a]	
Items	BC	NC	PC	T50	T100	T200	SEM	*P*	*L*	*Q*
Villus height, μm	293^{a}	131^{c}	263^{ab}	210^{b}	214^{b}	237^{b}	11	<0.001	<0.001	0.012
Crypt depth, μm	227^{b}	303^{a}	211^{b}	204^{b}	180^{b}	193^{b}	9	<0.001	<0.001	<0.001
V/C	1.31^{a}	0.43^{b}	1.28^{a}	1.03^{a}	1.20^{a}	1.23^{a}	0.06	<0.001	<0.001	<0.001

In the same row, values with no letter or the same letter superscripts mean no significant difference ($P > 0.05$), while with different letter superscripts mean significant difference ($P < 0.05$). n = 5. V/C, villus height/crypt depth; BC, blank control group, piglets without ETEC challenge were fed diet without supplements; NC, negative control group, piglets were fed diet without supplements and challenged by ETEC; PC, positive control group, piglets were fed diet supplemented with 50 mg kg^{-1} quinocetone and challenged by ETEC; T50, T50 group, piglets were fed diet supplemented with 50 mg kg^{-1} GE and challenged by ETEC; T100, T100 group, piglets were fed diet supplemented with 100 mg kg^{-1} GE and challenged by ETEC; T200, T200 group, piglets were fed diet supplemented with 200 mg kg^{-1} GE and challenged by ETEC.
[a] L, Linear; Q, Quadratic; Linear and quadratic effect of adding GE compared to the NC.

Table 6. Effect of GE on intestinal permeability of piglets

	Groups								Contrast[a]	
Items	BC	NC	PC	T50	T100	T200	SEM	*P*	*L*	*Q*
DAO, pg mL^{-1}	110^{b}	178^{a}	55^{d}	89^{bc}	68^{cd}	81^{c}	10	<0.001	<0.001	<0.001
ET-1, ng L^{-1}	217^{ab}	222^{a}	214^{b}	214^{b}	214^{b}	213^{b}	1	0.007	0.009	0.042
D-LA, μg L^{-1}	33.4^{b}	88.6^{a}	29.4^{b}	26.0^{b}	25.0^{b}	26.0^{b}	5.6	<0.001	<0.001	<0.001

In the same row, values with no letter or the same letter superscripts mean no significant difference ($P > 0.05$), while with different letter superscripts mean significant difference ($P < 0.05$). n = 5. D-LA, D-lactate; ET-1, endothelin-1; DAO, diamine oxidase; BC, blank control group, piglets without ETEC challenge were fed diet without supplements; NC, negative control group, piglets were fed diet without supplements and challenged by ETEC; PC, positive control group, piglets were fed diet supplemented with 50 mg kg^{-1} quinocetone and challenged by ETEC; T50, T50 group, piglets were fed diet supplemented with 50 mg kg^{-1} GE and challenged by ETEC; T100, T100 group, piglets were fed diet supplemented with 100 mg kg^{-1} GE and challenged by ETEC; T200, T200 group, piglets were fed diet supplemented with 200 mg kg^{-1} GE and challenged by ETEC.
[a] L, Linear; Q, Quadratic; Linear and quadratic effect of adding GE compared to the NC.

(Table 5). Dietary supplementation with 50–200 mg kg^{-1} GE linearly increased villus height (P < 0.001) and villus height to crypt depth ratio (P < 0.001), and decreased crypt depth (P < 0.001) in the jejunum of piglets compared with NC group (Table 5).

Intestinal permeability of weaned piglets was reflected by serum levels of diamine oxidase (DAO), D-lactate (D-LA) and endothelin-1 (ET-1). Compared with the BC group, ETEC challenge significantly increased (P < 0.05) the DAO, D-LA and ET-1 levels in the serum

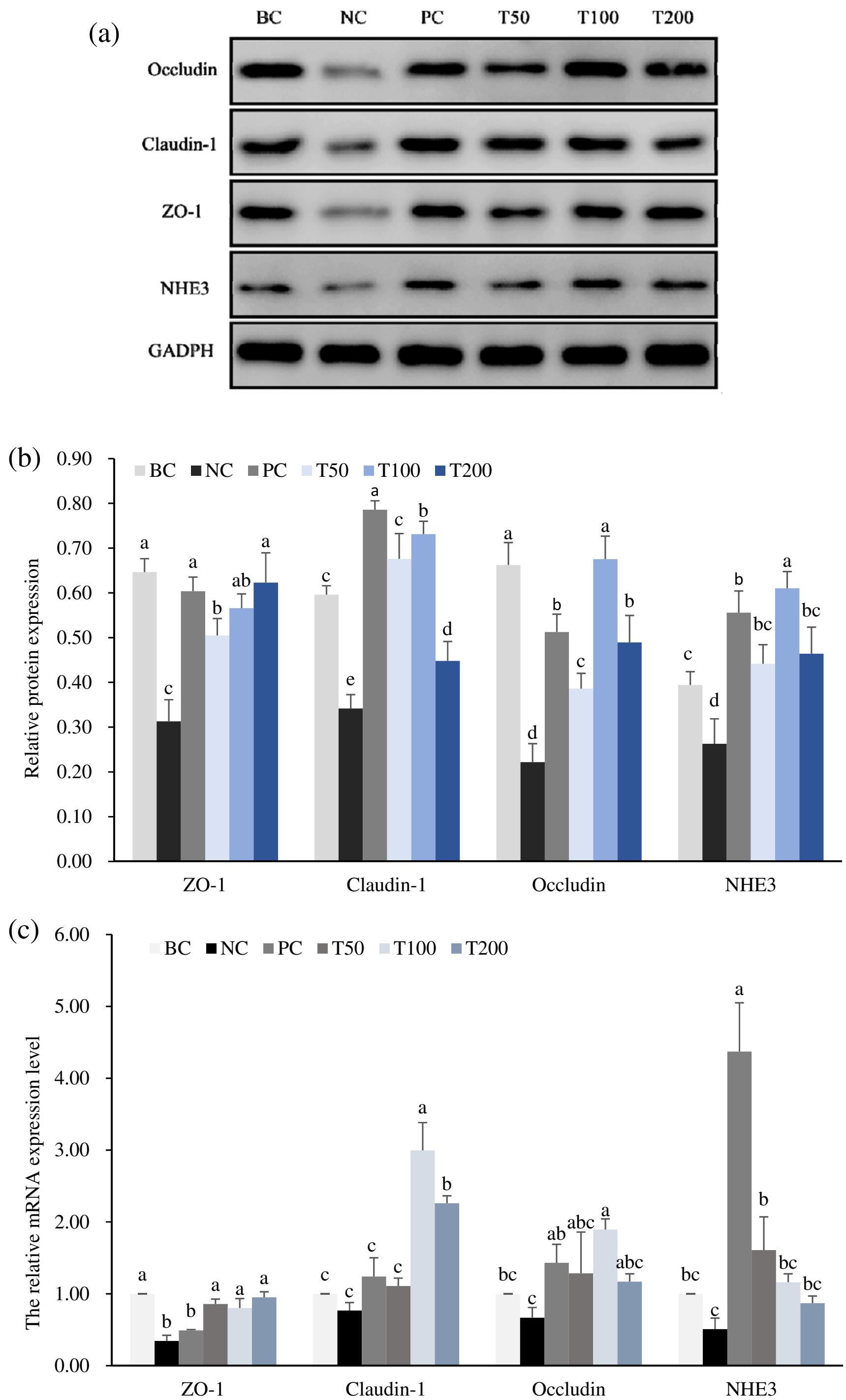

Figure 4. Effect of GE on tight junction expression in the jejunum mucosa of piglets. Protein expression of ZO-1, Occludin, Claudin-1 and NHE3 were determined by Western blotting (a). The results were normalized to the GAPDH standard (b). The relative mRNA expression of ZO-1, Occludin, Claudin-1 and NHE3 (c). Values are expressed as means ± SD (n = 5), and values with no common superscripts means significant difference (P < 0.05). BC, blank control group, piglets without ETEC challenge were fed diet without supplements; NC, negative control group, piglets were fed diet without supplements and challenged by ETEC; PC, positive control group, piglets were fed diet supplemented with 50 mg kg^{-1} quinocetone and challenged by ETEC; T50, T50 group, piglets were fed diet supplemented with 50 mg kg^{-1} GE and challenged by ETEC; T100, T100 group, piglets were fed diet supplemented with 100 mg kg^{-1} GE and challenged by ETEC; T200, T200 group, piglets were fed diet supplemented with 200 mg kg^{-1} GE and challenged by ETEC.

of piglets (Table 6). However, dietary supplementation with GE linearly reduced the levels of DAO ($P < 0.001$), D-LA ($P < 0.001$) and ET-1 ($P = 0.009$) in the serum of piglets compared with NC group.

Effect of dietary GE on tight junctions in the intestinal mucosa of piglets

As shown in Fig. 4, compared with the BC group, ETEC challenge significantly decreased the protein (Fig. 4(a) and (b)) and mRNA (Fig. 4(c)) expressions of ZO-1, Claudin-1, Occludin and NHE3 in the jejunum mucosa of piglets ($P < 0.05$). Dietary supplementation with quinocetone reversed the decreasing expression of ZO-1, Claudin-1, Occludin and NHE3 in the jejunum mucosa of piglets ($P < 0.05$) compared with NC group. Dietary supplementation with GE linearly increased the mRNA expression of ZO-1 ($P = 0.001$), Claudin-1 ($P = 0.003$), and linearly increased the protein expression of ZO-1 ($P < 0.001$), Claudin-1 ($P < 0.001$), Occludin ($P < 0.001$) and NHE3 ($P < 0.001$) in the jejunum mucosa of piglets compared with NC group.

DISCUSSION

In the chemical analysis, we found that phenolics were the principal constituents of GE and they were usually considered nontoxic due to their ubiquity in many sorts of plant-derived foods and beverages. The structure of phenolic compounds highly influences their mechanisms of pharmacological action. Some studies have proved that antimicrobial and antioxidant activities of phenolics were related to their aromatic structure and multiple hydroxyl groups, since it can provide electrons or hydrogen atoms to neutralize free radicals and other reactive oxygen species (ROS).[15,16] Moreover, flavonoids, such as catechin (**3**), epicatechin (**4**), naringenin (**32**), kaempferol (**34**), morin (**20**) and quercetin (**30**), *etc*., have been also described as having antimicrobial and antioxidant activities.[17,18]

Enterotoxingenic *Escherichia coli* (ETEC) is a major cause of diarrhea in neonatal and weaned piglets,[19] which can lead to intestinal barrier dysfunction, inflammation, and accompanied by a reduction in the absorption capacity of the small intestine.[14,20] Our current results are consistent with previous research, which administration of ETEC caused diarrhea in piglets. Therefore, appropriate nutrition intervention appears to be essential for enhancing the gut health of piglets. In the present study, dietary addition of GE could reduce the diarrhea incidence of piglets compared with the NC group. It has been reported that phenolics in GE, such as flavonoids and tannins, presented anti-diarrheal activity due to enable to denature protein and form protein complex to reduce the intestinal mucosa permeability.[21] Additionally, quercetin and its derivatives in the GE might be the main contributing factors of antidiarrheal activity. Hirudkar *et al*. reported that quercetin could inhibit the intestinal secretion along with reactivation of Na^+/K-ATPase activity in rats, and then relieve diarrhea of rats induced by enteropathogenic *Escherichia coli*.[10] However, compared with the BC group, administration of ETEC or dietary GE addition did not cause the significant changes in ADFI, ADG and F/G of piglets. We observed that diarrhea mainly occurred in the first 5 days after ETEC challenge, and there was no obvious diarrhea on the 10th day after ETEC challenge, which was consistent with the results of previous studies.[22,23] In other words, ETEC challenge reduced the growth performance of piglets in the weaned piglet model might mainly during 10 days after ETEC challenge. Kim *et al*. found that pigs challenged by ETEC had lower ADG and ADFI from day 0 to day 5 post inoculation compared with pigs without ETEC challenge. They also found there were no differences were observed in ADG, ADFI, and F/G among groups from day 5 to day 11 post inoculation of ETEC.[22]

Previous studies have shown that the growth performances, as well as the diarrhea incidence are associated with oxidative stability and intestinal health status.[24] We analyzed the antioxidative index in the serum of piglets. The results showed that dietary addition with 50–200 mg kg^{-1} GE could improve the antioxidant capacity of piglets. The effect of GE on the antioxidative capacity is consistent with previous studies,[8,25,26] which mainly contribute to its rich flavonoids and other phenolics.[27] Antioxidant activities of flavonoids and phenolics are related to their aromatic structure and multiple hydroxyl groups, since they can act as hydrogen donors to provide electrons or hydrogen atoms, and neutralize free radicals and scavenge reactive oxygen.[8]

Usually, administration of ETEC could induce immune activation and uncontrolled inflammatory,[28,29] which leads to an imbalance between the pro- and anti-inflammatory cytokines. Proinflammatory cytokines, such as TNF-α, IL-6, and IL-1β, have been showed to mediate the host inflammatory process and initiate an effective inflammatory response to pathogen infection.[30] The current research shown that ETEC challenge increased the serum concentration and mRNA expression of TNF-α, IL-6, and IL-1β in the jejunum mucosa in weaned piglets, which were thought to reflect the host's inflammatory response to the ETEC invasion.[31,32] This inflammatory response was attenuated by dietary GE supplementation. Dietary supplementation with 50–200 mg kg^{-1} of GE decreased serum levels and jejunum mucosa mRNA expression of TNF-α, IL-6, and IL-1β of piglets challenged by ETEC. Jiang *et al*. have reported that GE could significantly inhibit IL-6 production in the RAW264.7 macrophages induced by lipopolysaccharide (LPS).[33] They found that GE could suppress LPS-induced iNOS expression through suppression of the ERK 1/2 MAPK signaling pathway, and then inhibited NO production in LPS-induced RAW264.7 cell.[33] NO, which is synthesized by iNOS, is a well-known proinflammatory mediator that is involved in various physiological and pathological process.[34] Thus, GE supplementation can relieve intestinal inflammation in piglets challenged by ETEC might through reducing the secretion of inflammatory mediator and suppression of the inflammation signaling pathway.

Infection with ETEC release enterotoxins to impair the intestinal barrier function, and change intestinal permeability, which indirectly increase the secretion of water and electrolytes into the intestinal lumen, resulting in secretory diarrhea.[35] Thus, we further evaluated the effect of dietary GE on the intestinal morphology and permeability of piglets challenged by ETEC. In the present study, piglets supplemented with GE in their dietary had higher villus height and villus height to crypt depth ratio in the jejunum than piglets in NC group, which suggested that GE could prevent the damage of intestinal morphology caused by ETEC infection. The DAO and D-LA contents are recognized as sensitive markers for monitoring the alteration of intestinal barrier permeability.[36] DAO is expressed mainly in the small intestine and rarely in the serum under normal circumstances, and serum DAO levels will increase when intestinal barrier integrity is damaged.[37] Similarly, when the balance of microbiota in the intestine is broken, pathogens multiply and produce excessive amounts of D-LA that are released into the blood through the damaged mucosa. Also, ET-1 is an enterotoxin released by microbiota in the intestine, and will release into the blood through the impaired intestinal barrier.[38] In the current study, the challenge of ETEC induced more DAO, D-LA and ET-1 contents in serum of piglets, and GE supplementation alleviated the increase, which was consistent with the lower diarrhea incidence compared with infected piglets in NC.

Intestinal endotoxin enters the blood circulation through non-specific paracellular or transcellular transport pathways when tight junction protein complexes were impaired and integrity of the intestinal barrier was reduced.[39] Tight junctions are important components of the intestinal mucosal barrier preventing the paracellular diffusion of intestinal bacteria across the epithelium,[40] which including zonula occludens, occludin, and claudins. Na^+/H^+ exchanger (NHE) isoforms play important roles in intracellular pH regulation and in fluid absorption.[41] The isoform NHE3 localized to surfaces of epithelia and facilitate the absorption of NaCl.[42] Infection with ETEC impaired the intestinal barrier function, and reduced the absorption of water and electrolytes from the intestinal lumen to tissues, and then resulting in watery diarrhea.[43] Therefore, the reduced expression of tight junction proteins and NHE3 were also in agreement with the increased diarrhea incidence, mucosal damage and circulation endotoxin levels in the piglets challenged by ETEC in the present study. However, GE have shown to ameliorate intestinal permeability and mucosal damage, and enhance tight junction integrity, giving a promising avenue for preventing and protecting the barrier function of the intestine. Anti-diarrhea activity of GE is mainly related to its content of flavonoids and tannins, which through denaturing protein hence forming interaction protein-tannates to reduce the intestinal mucosa permeability.[44,45] Meanwhile, flavonoid and phenolic in GE have a good antibacterial effect on intestinal bacteria,[46] such as *Salmonella* and *Esherichia coli*.,[47,48] and then could alleviate diarrhea of piglets challenged by ETEC.

CONCLUSION

Collectively, our study demonstrated that GE could effectively alleviate diarrhea, ameliorate intestinal permeability and mucosal damage, and enhance tight junction integrity of intestine of piglets challenged by ETEC. The biological activity of GE is might mainly attributed to its active ingredients, such as flavonoids and phenolics, *etc*.

ACKNOWLEDGEMENTS

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AUTHOR CONTRIBUTIONS

Dingfa Wang and Hanlin Zhou designed the experiment. Luli Zhou analyzed the compositions of GE. Dingfa Wang analyzed the data and wrote the paper. Hu Haichao and Hou Guanyu finished the animal experiments. All authors reviewed and approved the final manuscript.

CONFLICT OF INTERESTS

The authors declared that there was no conflict of interests regarding the publication of this article.

ETHICS APPROVAL

All animal experiments were conducted in accordance with the Laboratory Animal Requirements of Environment and Housing Facilities and approved by the Institutional Animal Care and Use Committee of the Chinese Academy of Tropical Agricultural Sciences (Haikou, China).

SUPPORTING INFORMATION

Supporting information may be found in the online version of this article.

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